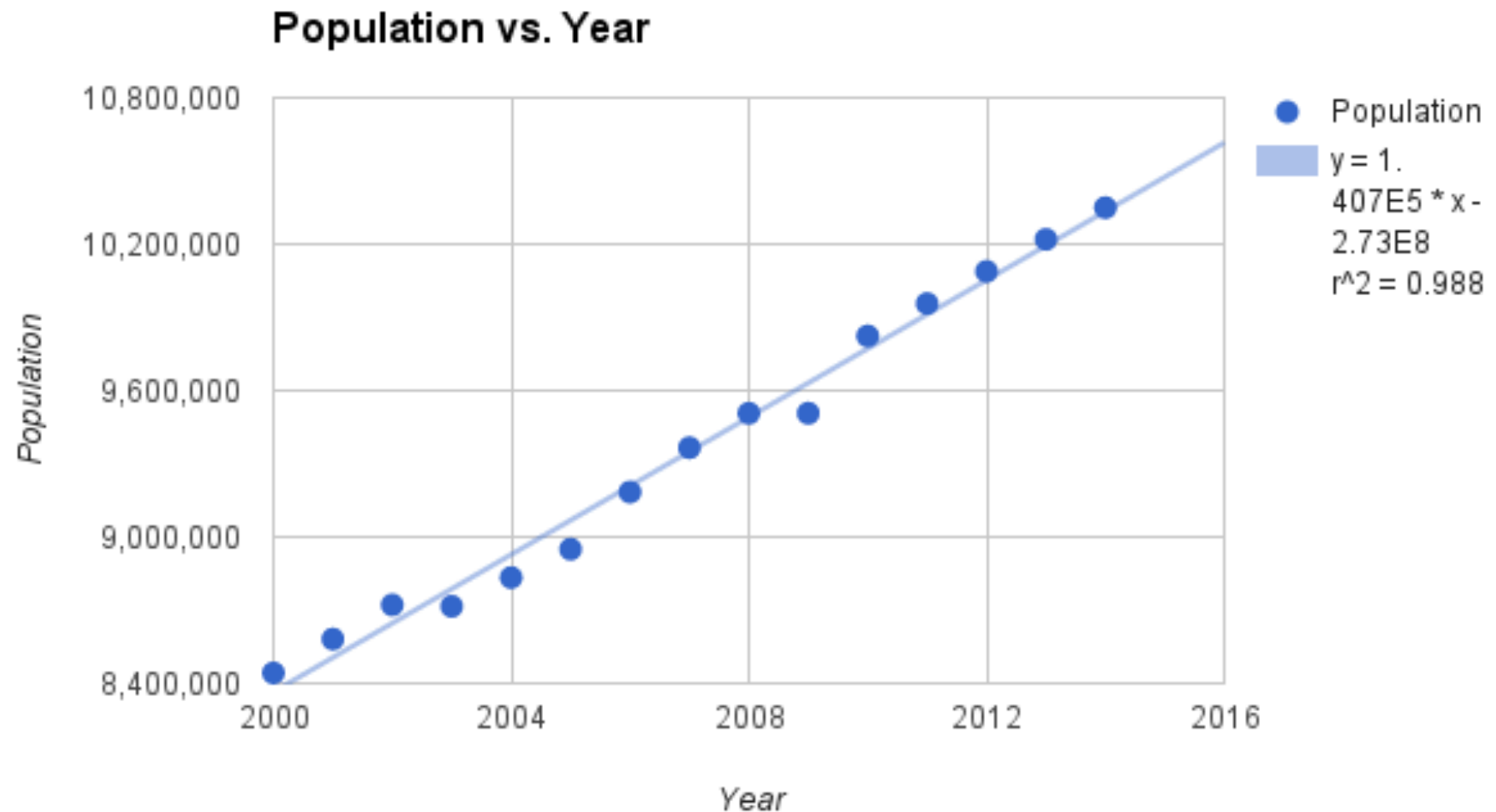


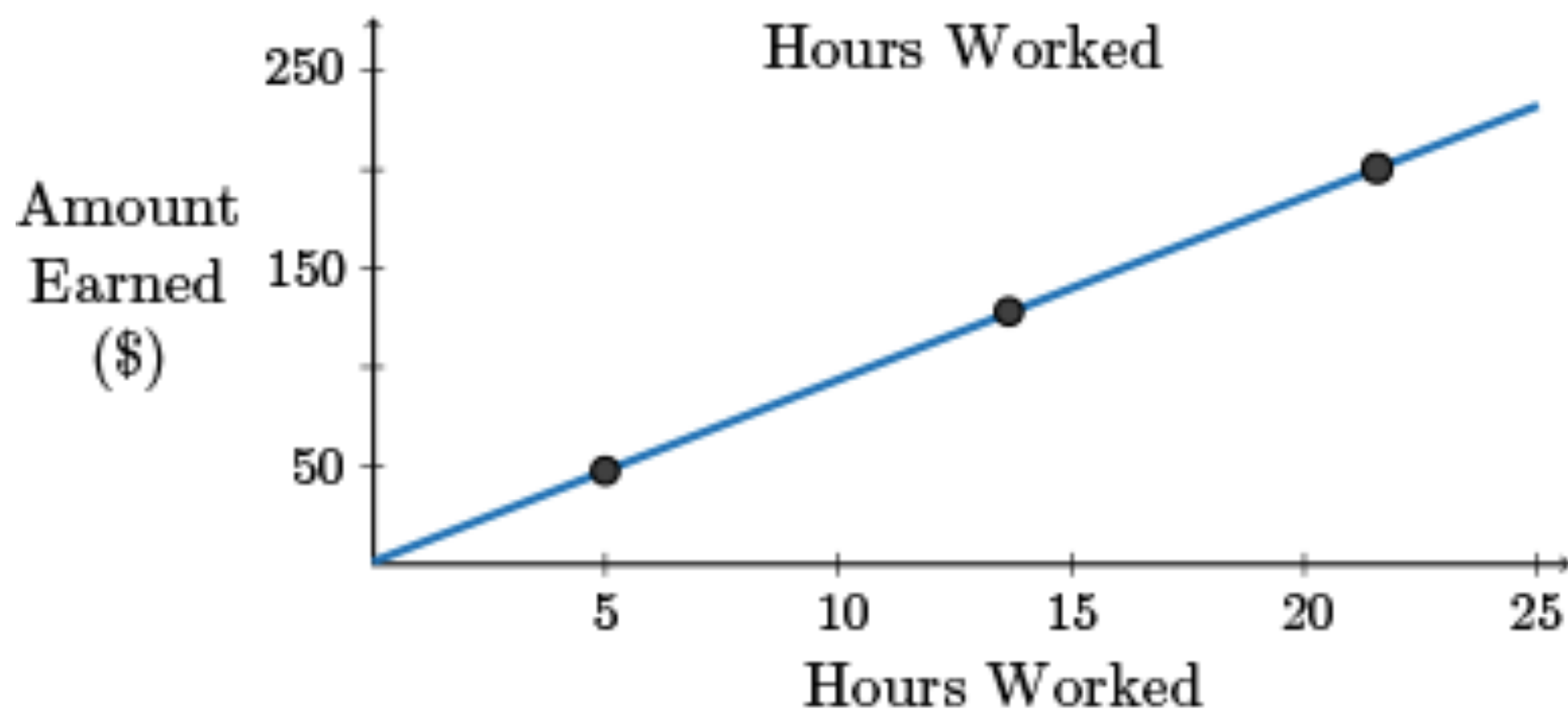
Linear Relationships

Introduction

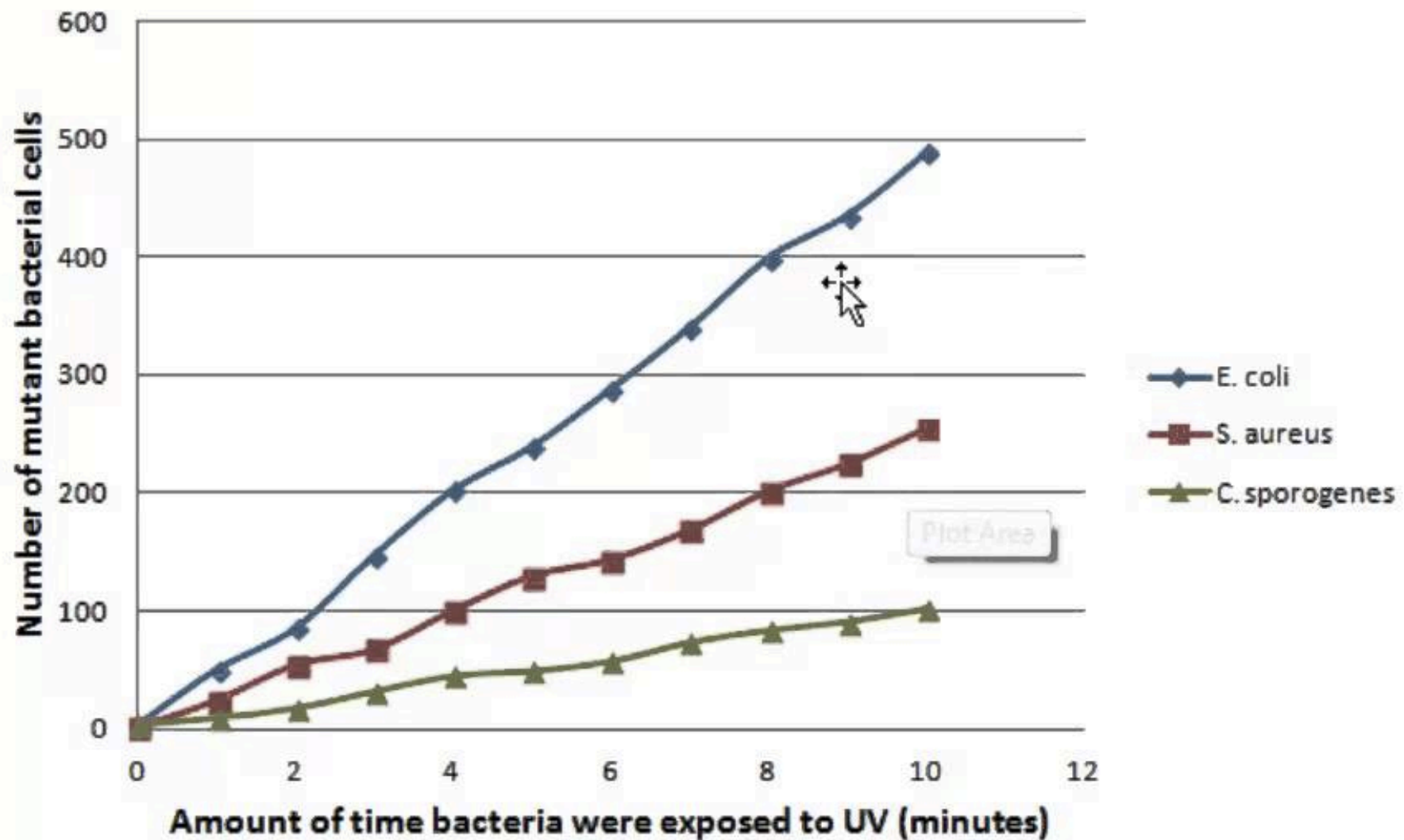
Population



Amount Earned Versus
Hours Worked



The effect of UV exposure on the appearance of mutant cells in 3 different bacterial populations



What is gradient?

Gradient is the measure of how **steep** a line is. The bigger the gradient, the steeper the line.

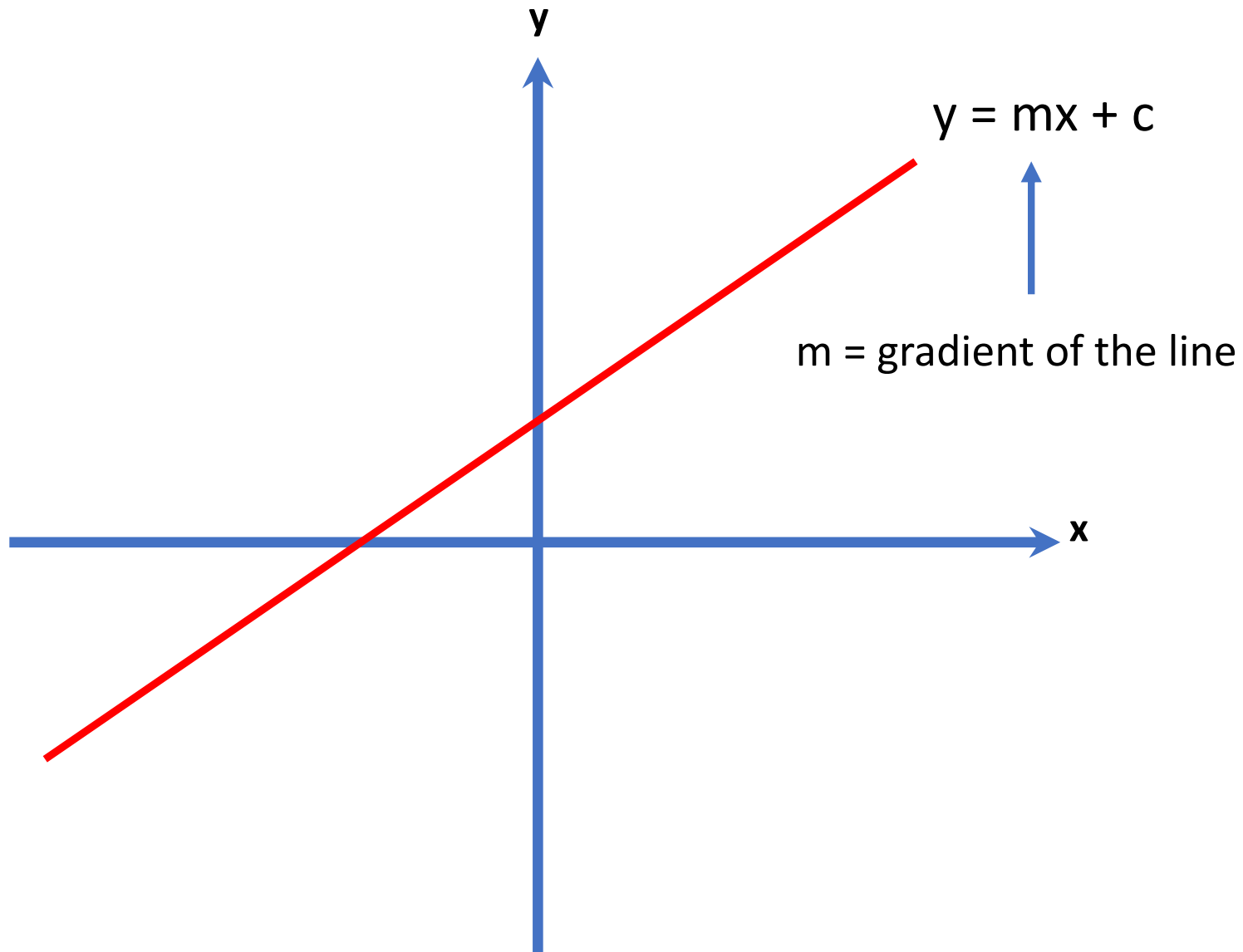
Small gradient

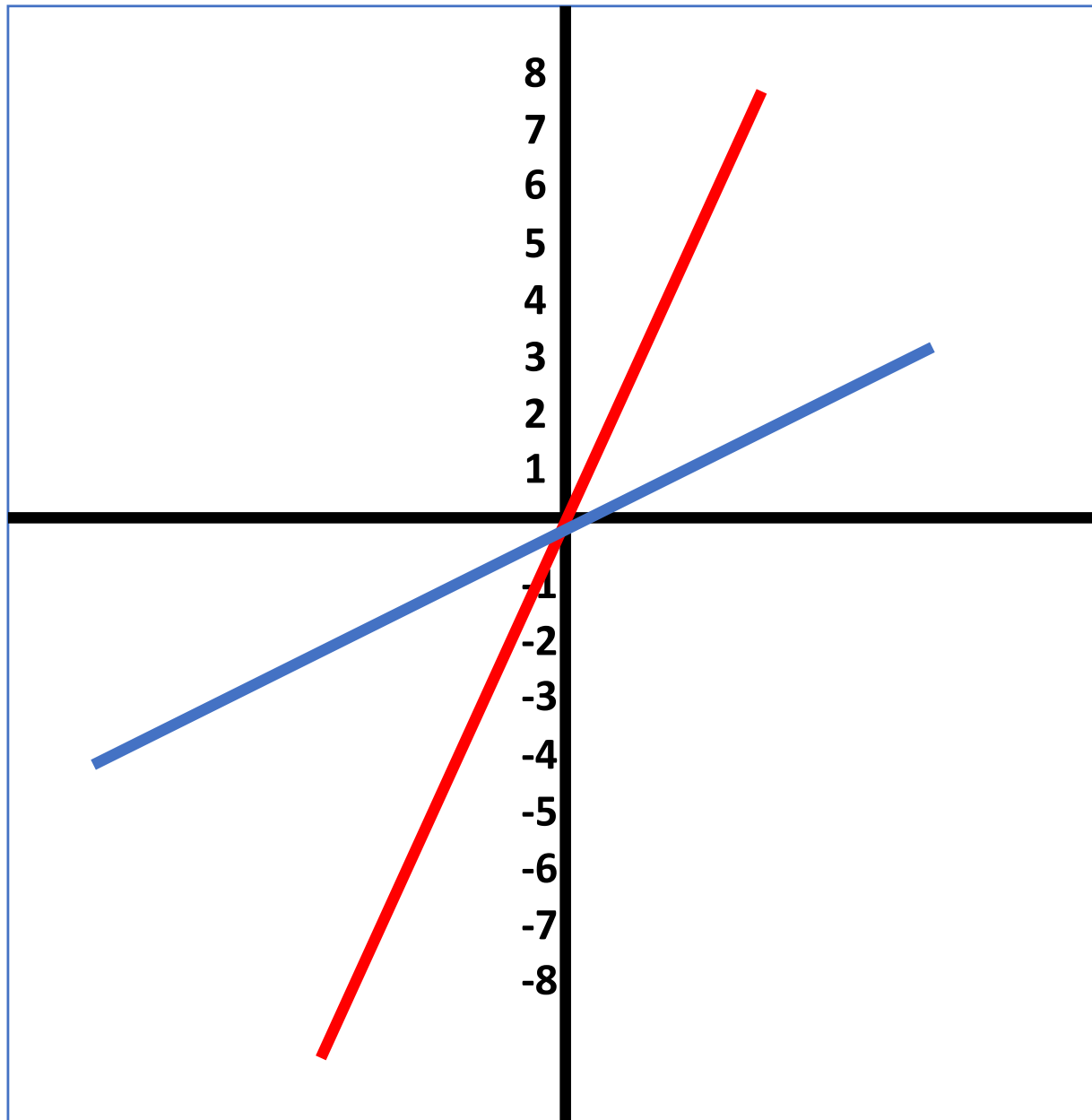


Large gradient



Consider a linear relationship

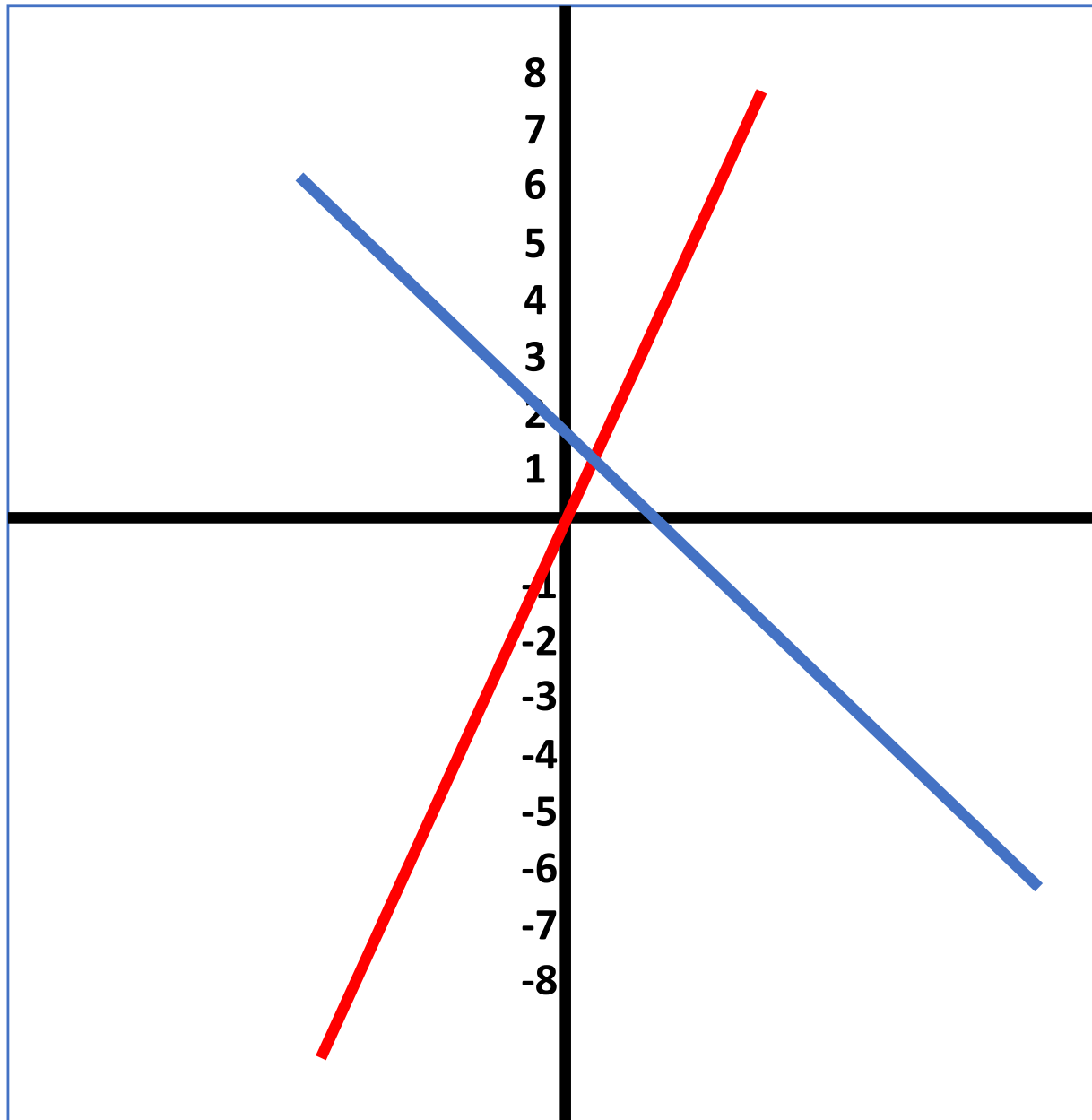




The **RED** line has a larger gradient than the **BLUE** line.

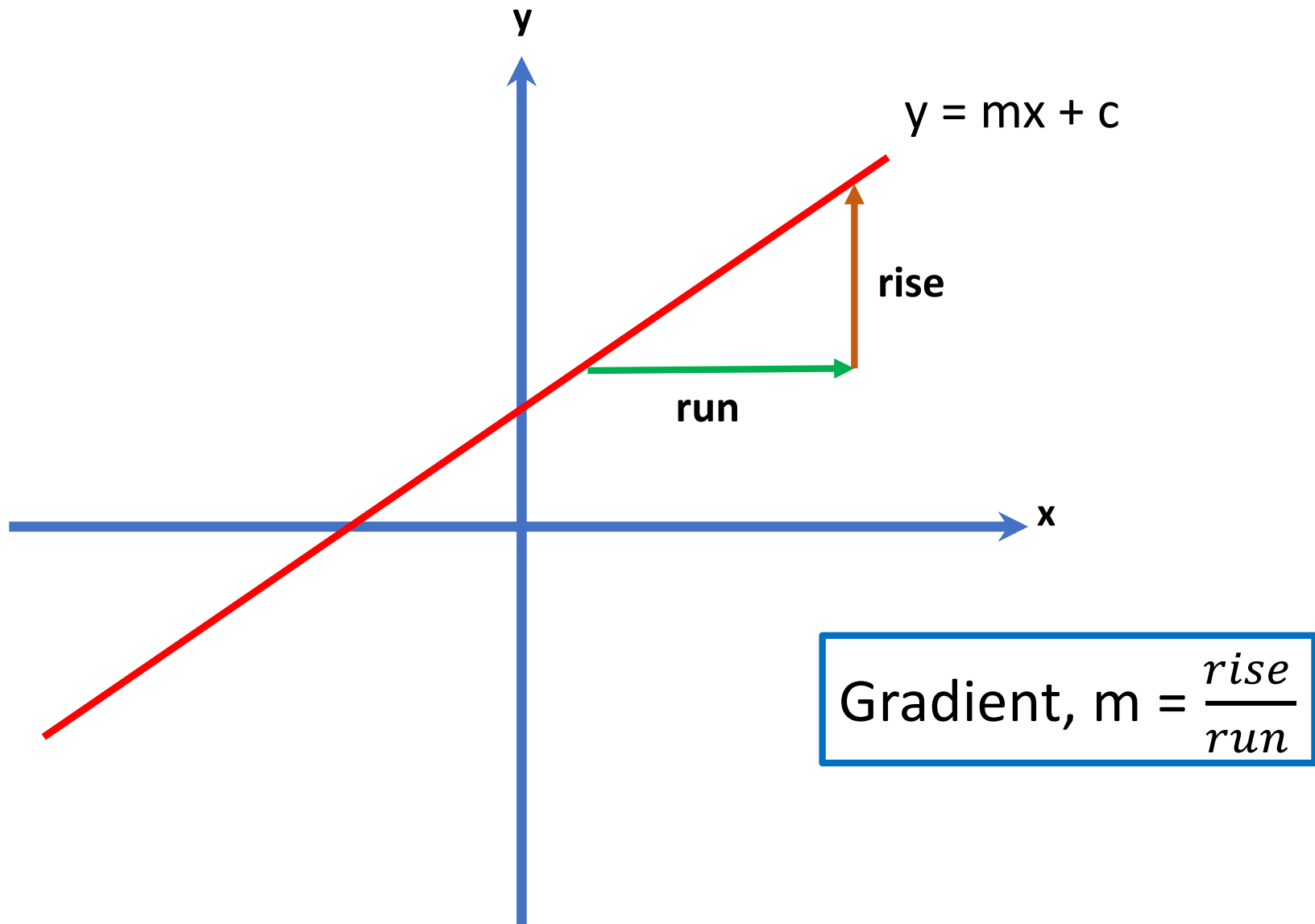
A big gradient gives a STEEP line

A small gradient gives a SHALLOW line

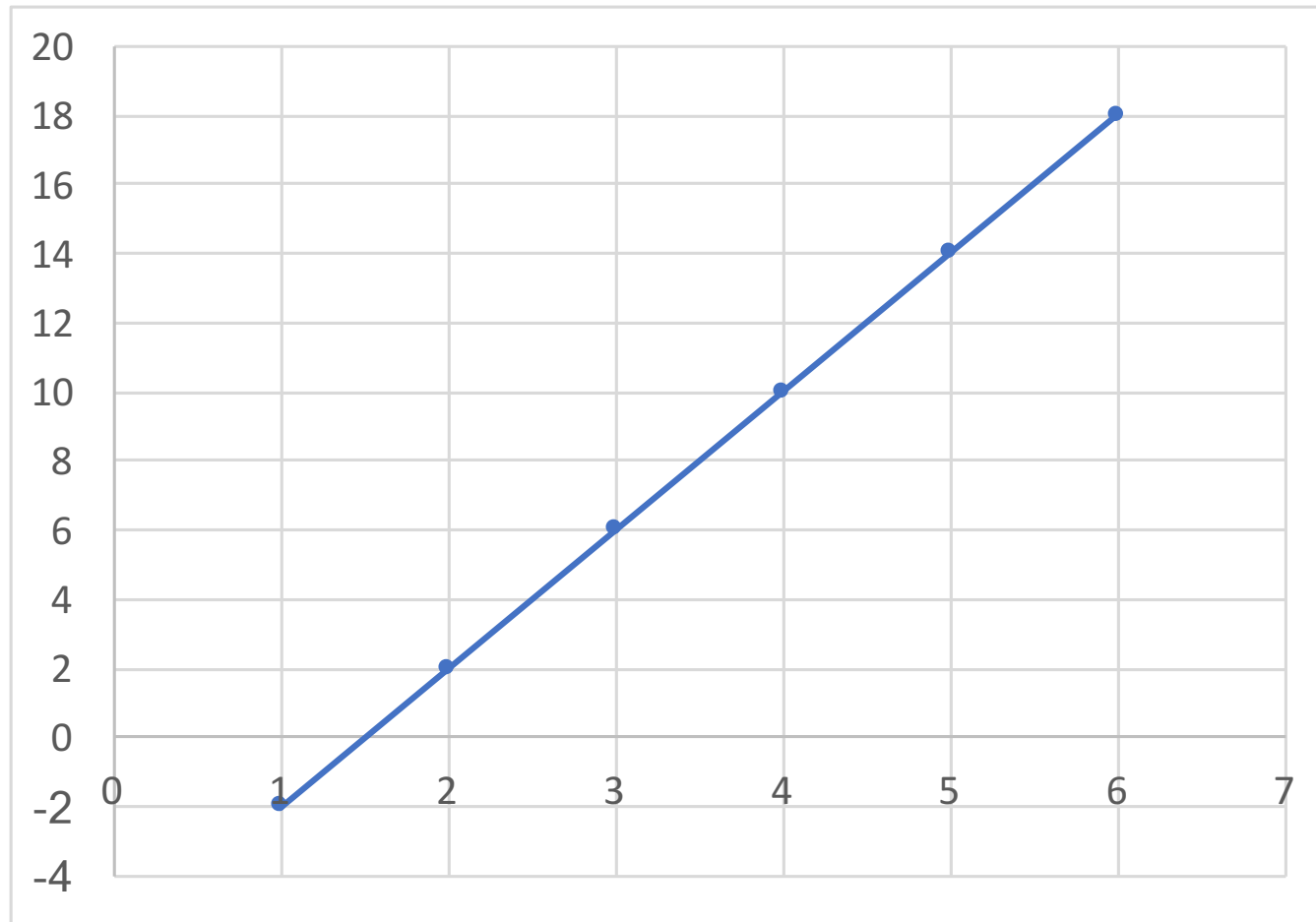


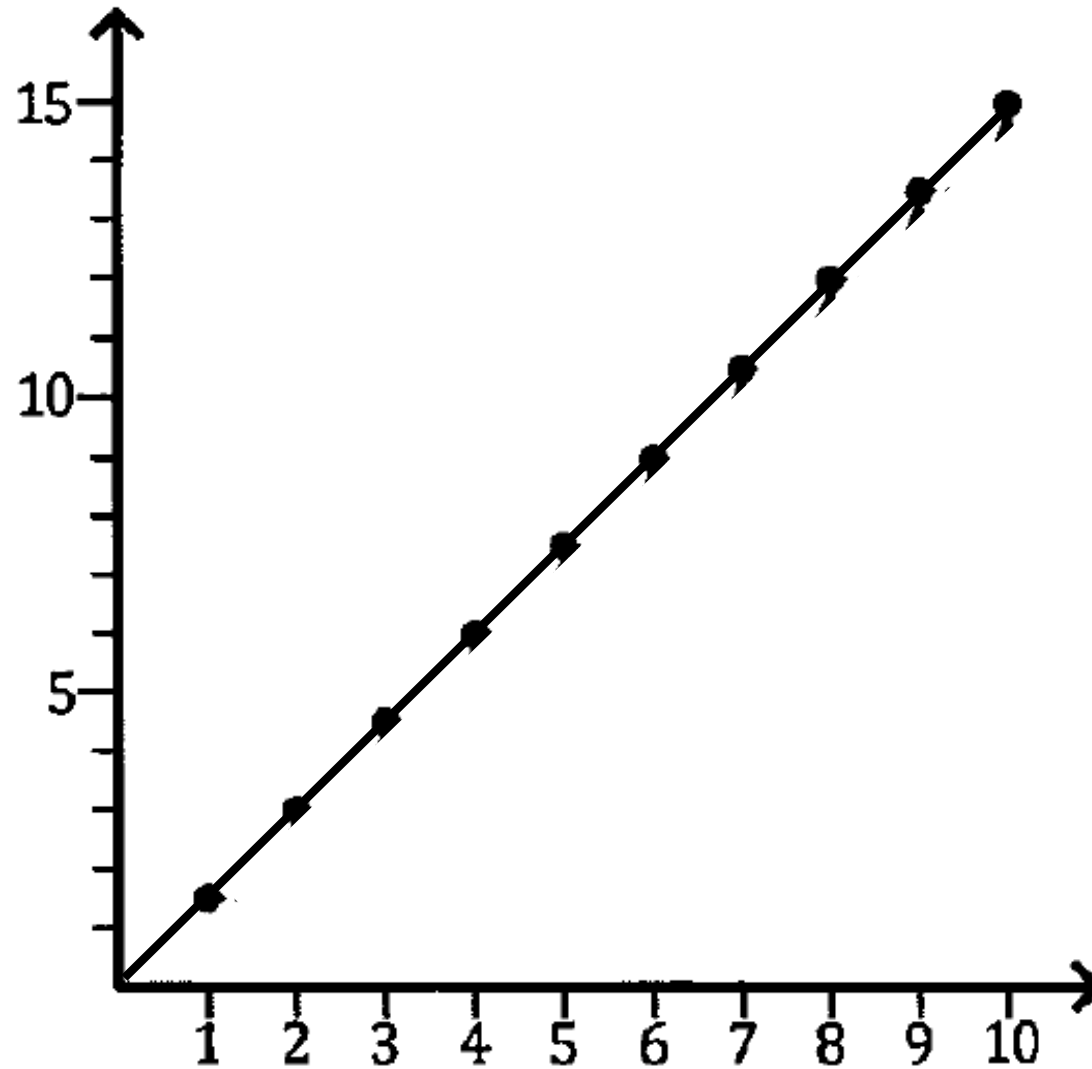
The **RED** line has a positive gradient, the **BLUE** line has a negative gradient

How do you define the gradient?



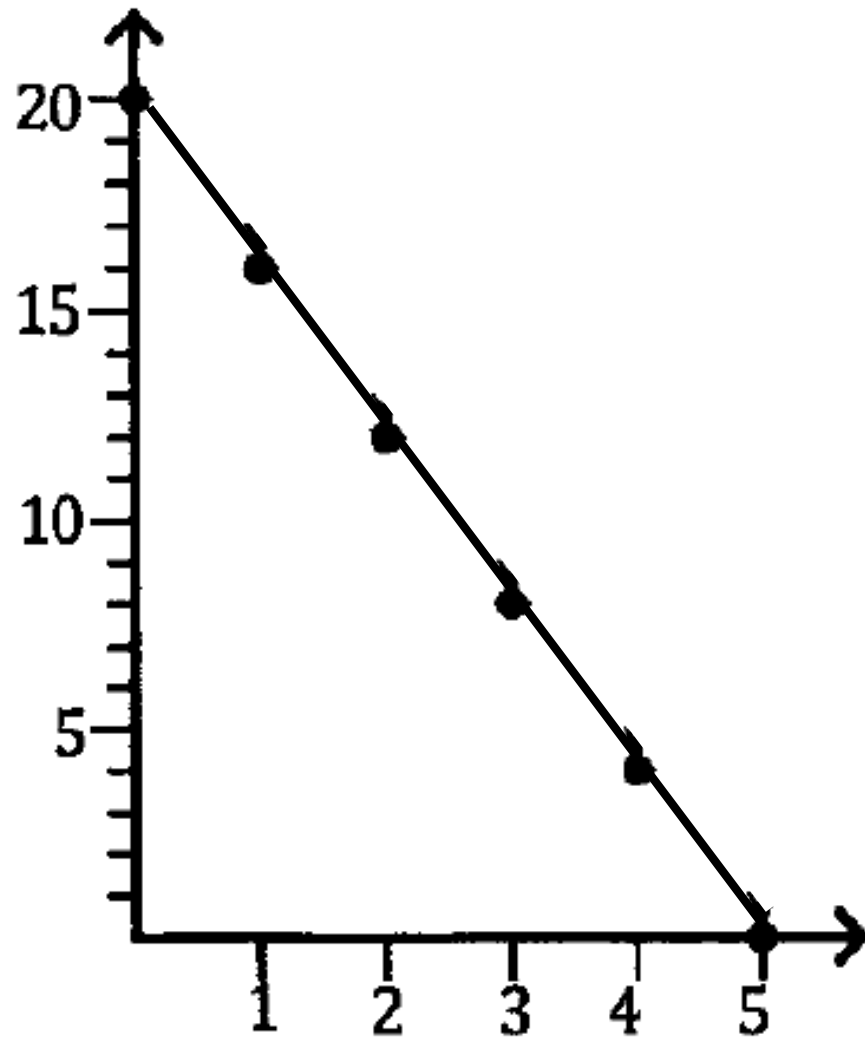
x	1	2	3	4	5	6
y	-2	2	6	10	14	18





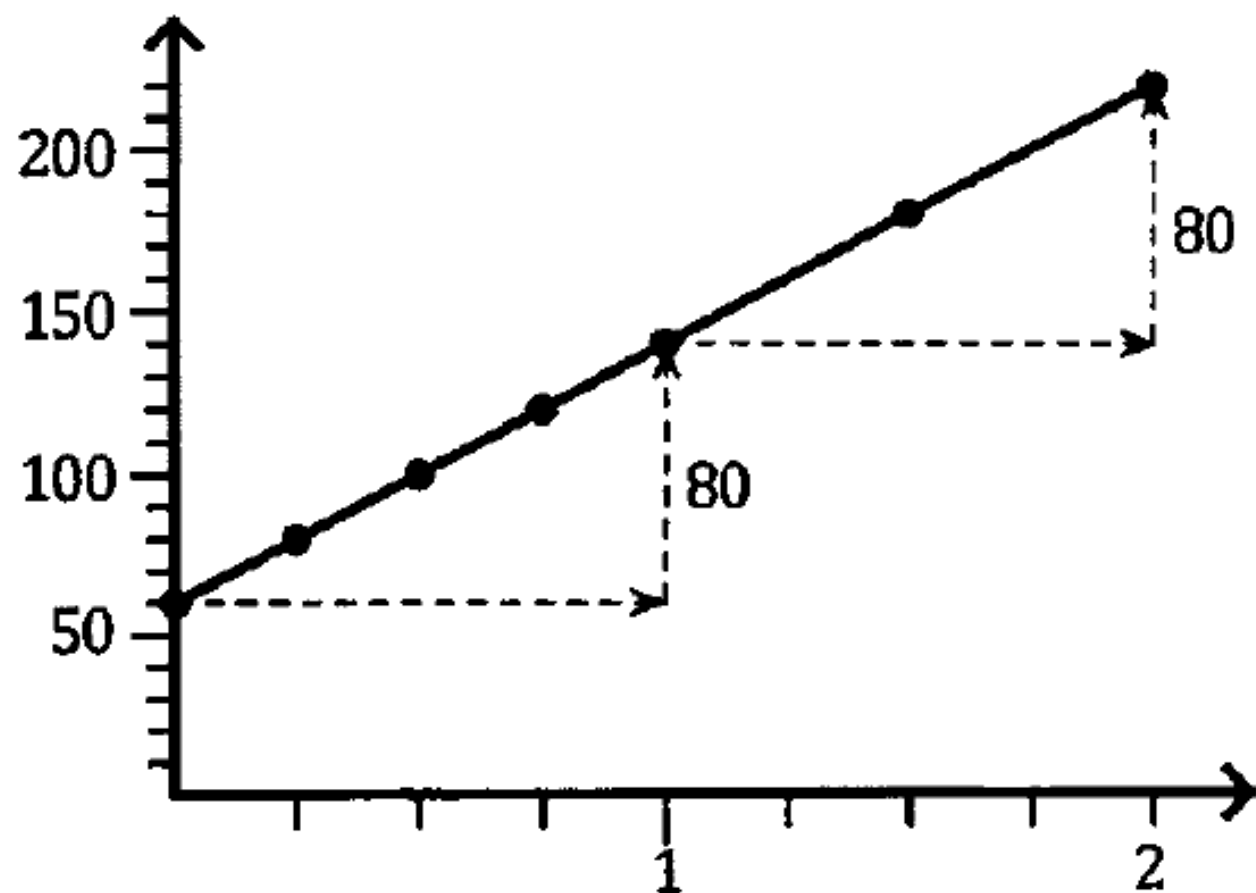
Each time we move
right 1 unit we move
up 1.5 units.

Gradient = 1.5



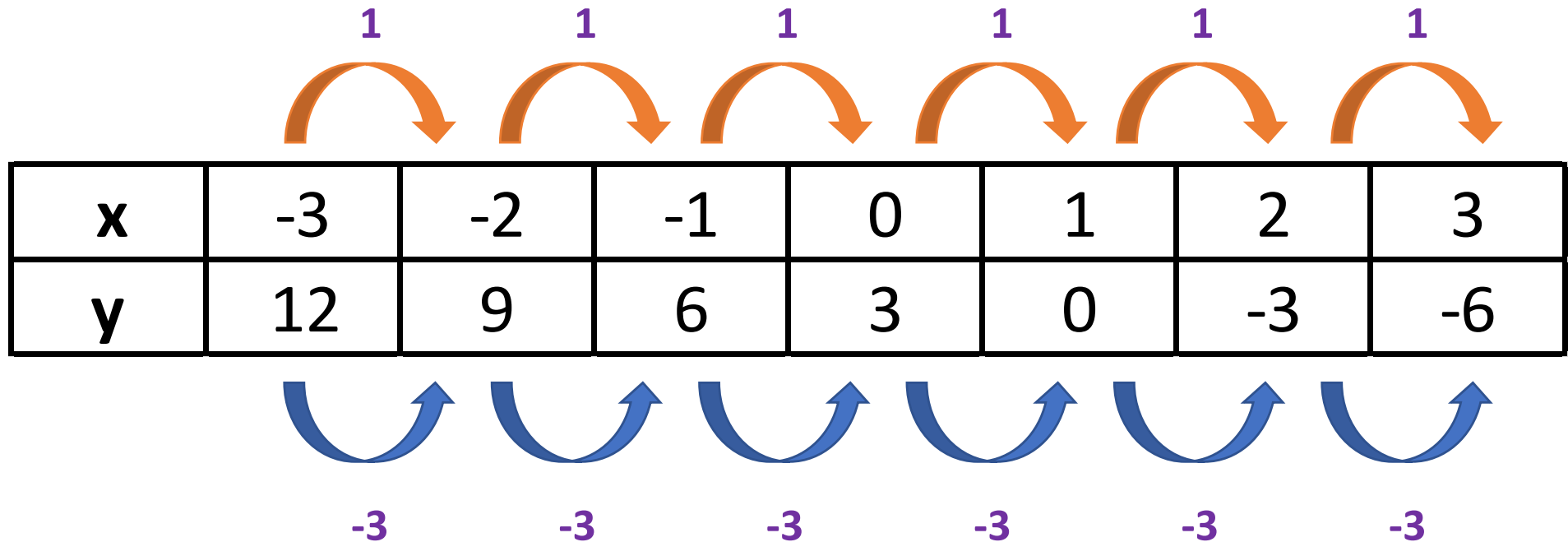
Each time we move
right 1 unit we move
down 4 units

$$\text{Gradient} = -4$$



Each time we move
right 1 unit we move
up 80 units.
(Moving right one quarter of a
unit sees a vertical rise of just 20
units.)

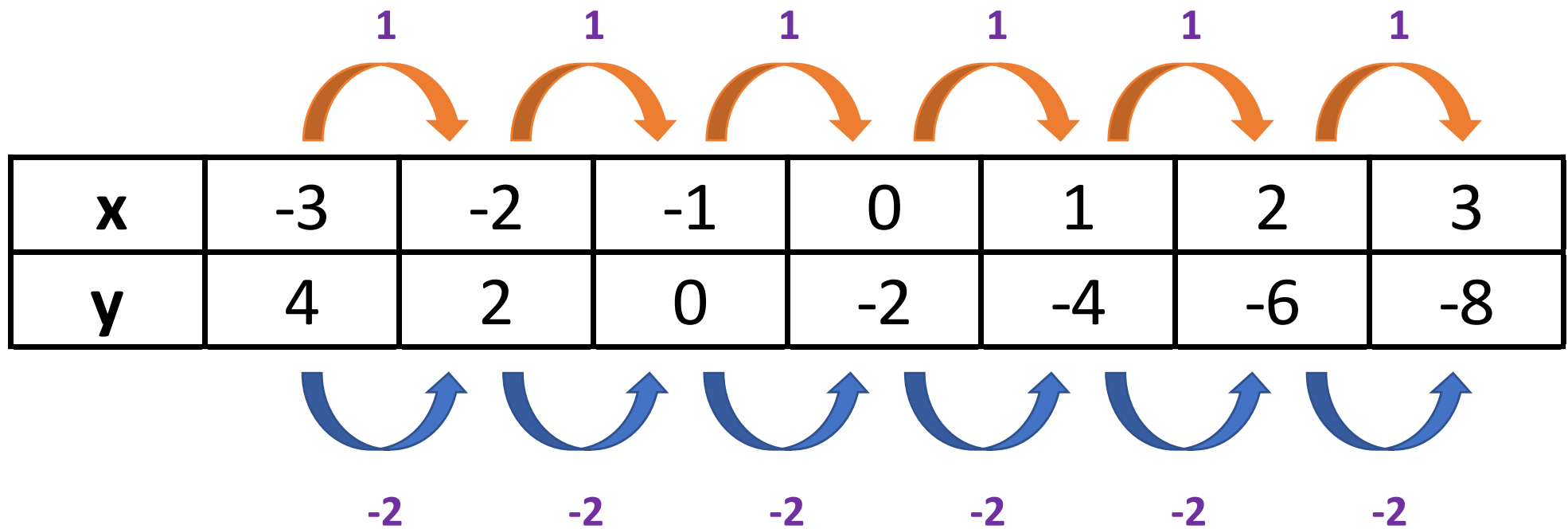
Table of values



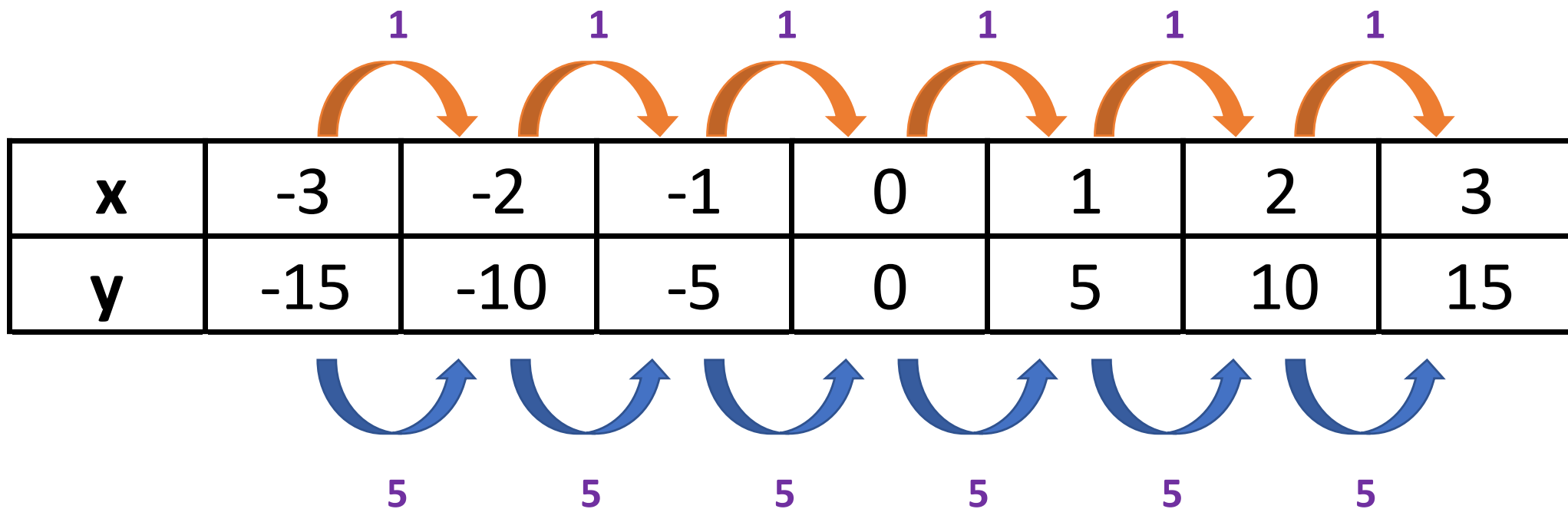
The diagram illustrates a linear function using a table of values. The table has two rows: 'x' and 'y'. The x-values range from -3 to 3, and the y-values range from 12 to -6. Above the table, orange curved arrows point from each x-value to the next, with a purple '1' above each arrow, indicating a constant difference of 1 in x. Below the table, blue curved arrows point from each y-value to the next, with a purple '-3' below each arrow, indicating a constant difference of -3 in y.

x	-3	-2	-1	0	1	2	3
y	12	9	6	3	0	-3	-6

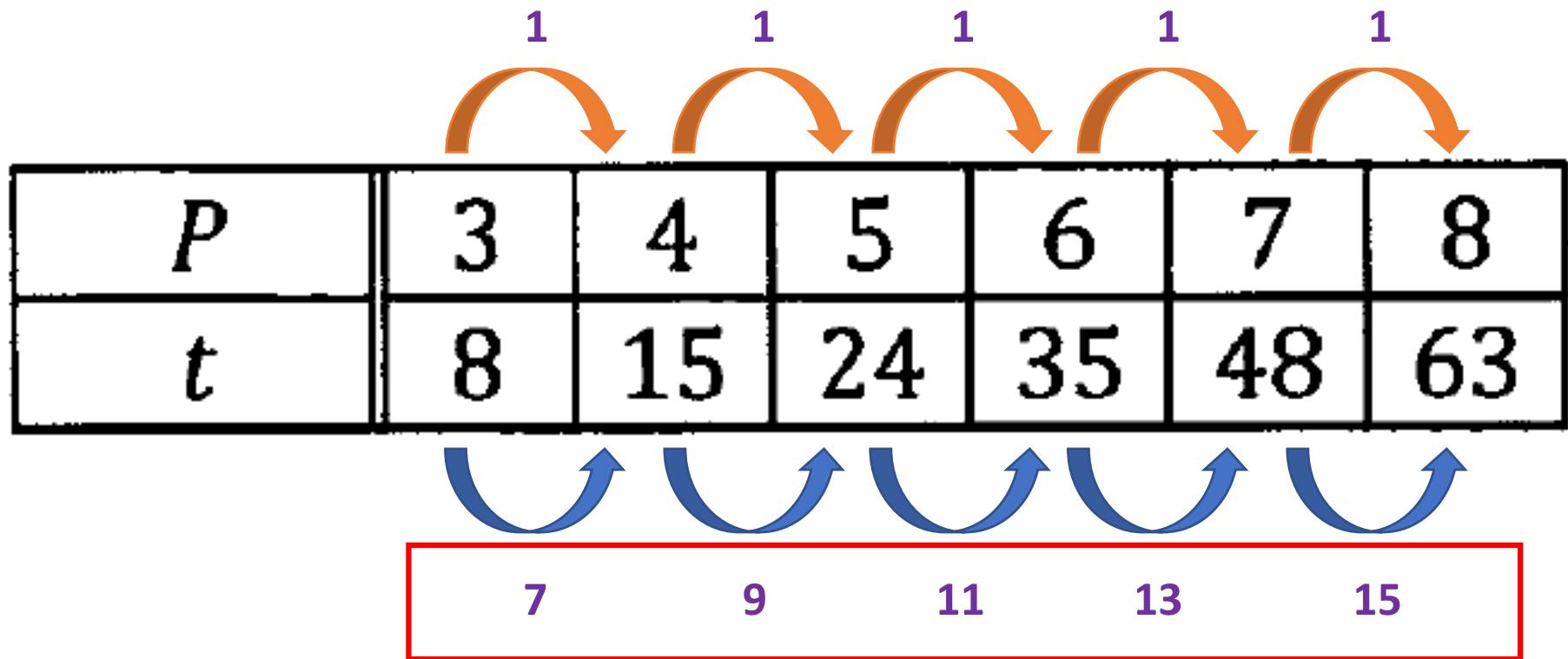
The difference = the gradient, m
Therefore in this case, $m = -3$



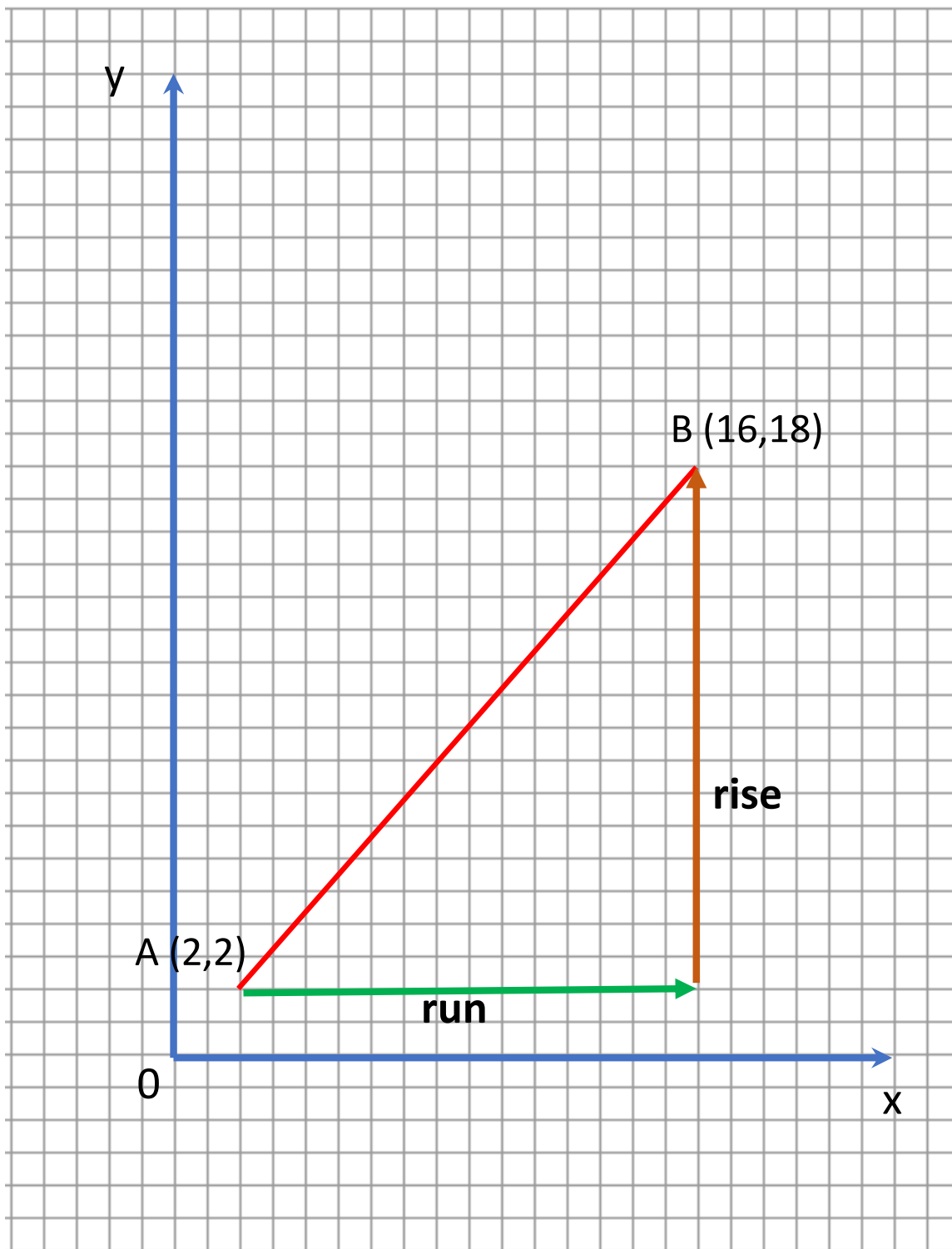
The difference = the gradient, m
Therefore in this case, $m = -2$



The difference = the gradient, m
Therefore in this case, $m = 5$



Difference is not constant, therefore this graph is not a linear graph.



$$\text{Gradient} = \frac{\text{rise}}{\text{run}}$$

Run is the change in **x** values when we move from A to B. **[HORIZONTAL]**

Rise is the change in **y** values when we move from A to B. **[VERTICAL]**

In the case,

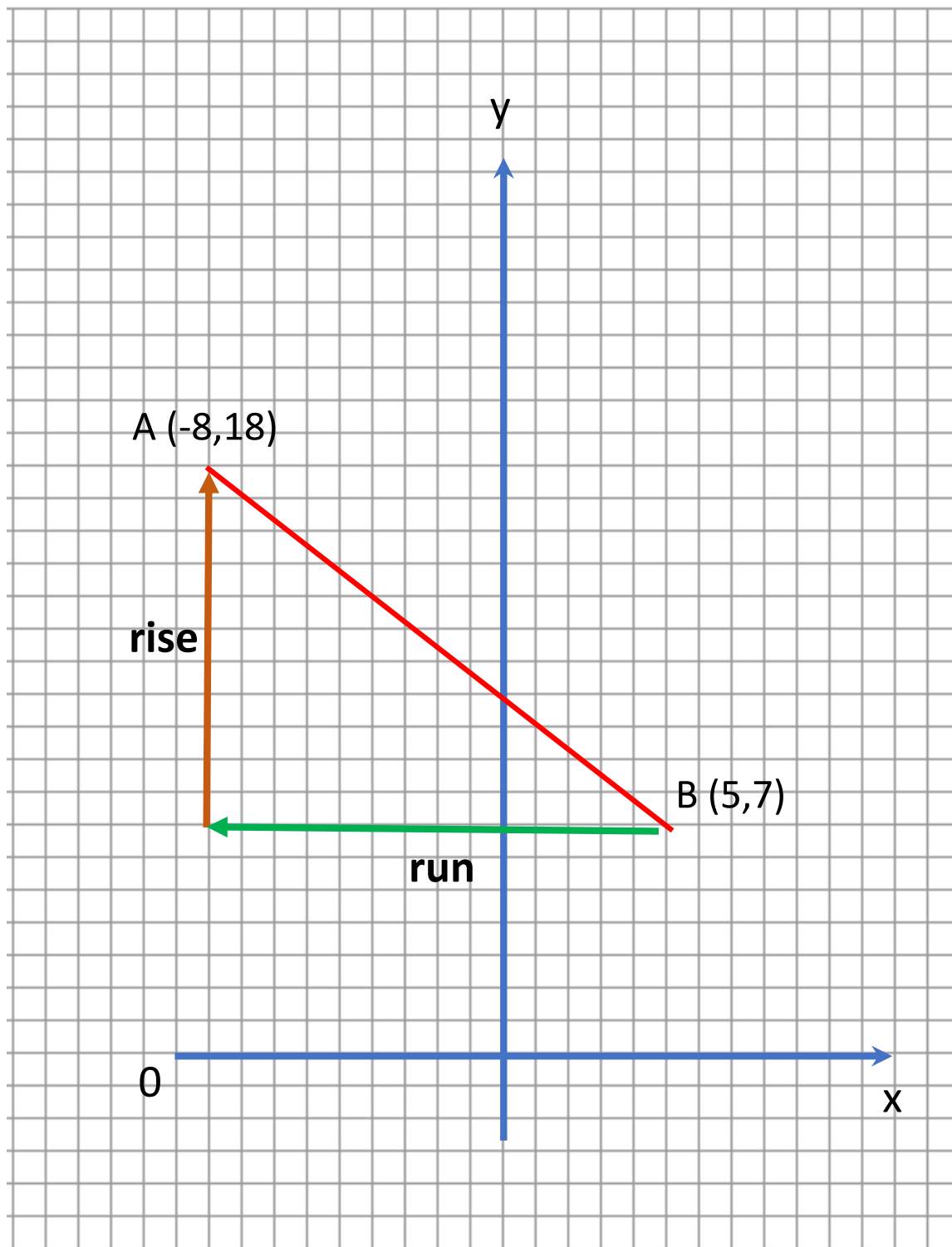
$$\text{Run} = 16 - 2 = 14$$

$$\text{Rise} = 18 - 2 = 16$$

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{16}{14} = \frac{8}{7}$$

As we move from A to B, the y value increases as the x value increases.

This gradient is **positive**.



In the case, the gradient for the line AB

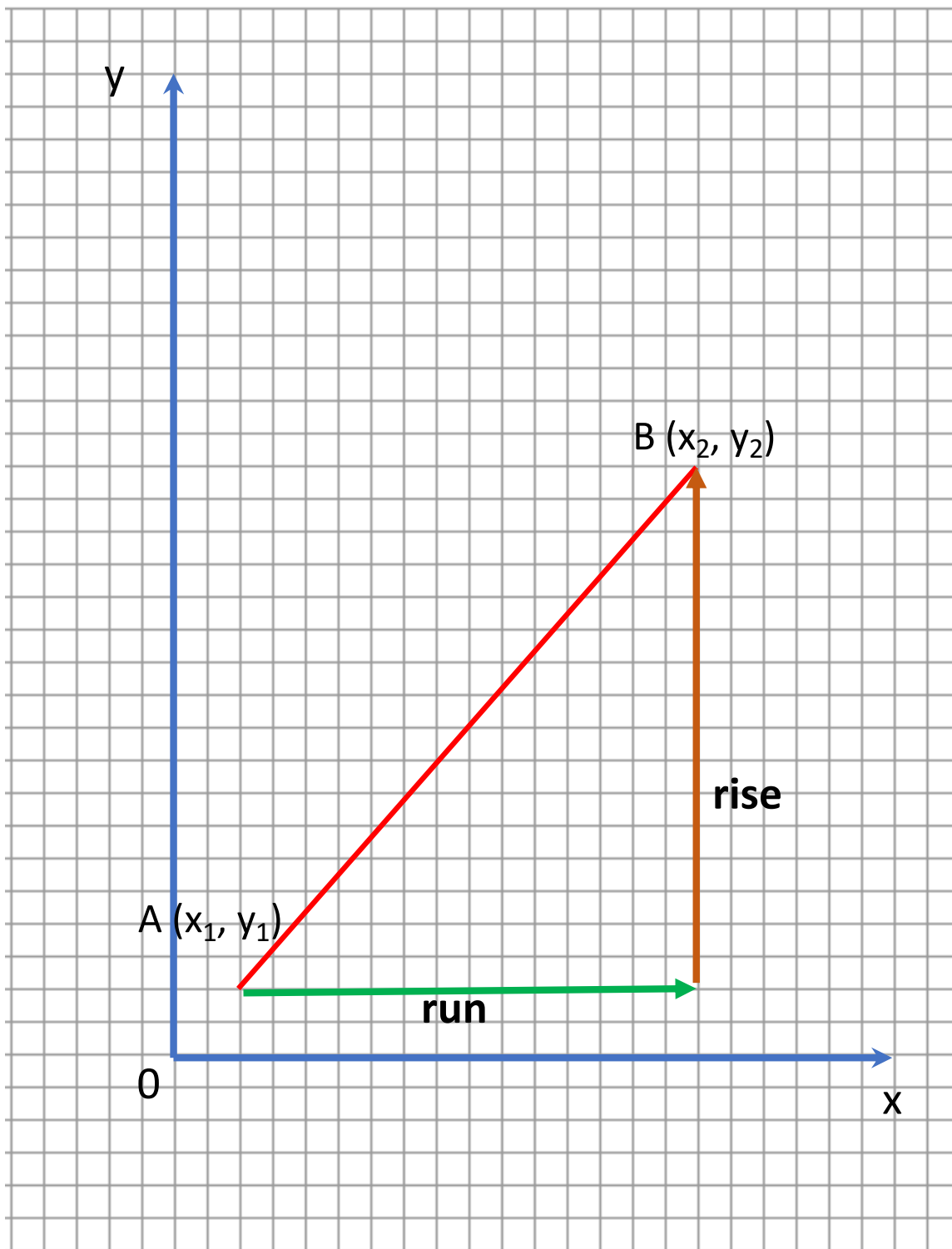
$$\text{Run} = 5 - (-8) = 13 \text{ [HORIZONTAL]}$$

$$\text{Rise} = 7 - 18 = -11 \text{ [VERTICAL]}$$

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = -\frac{11}{13}$$

As we move from A to B, the y value decreases as the x value increases.

This gradient is *negative*.



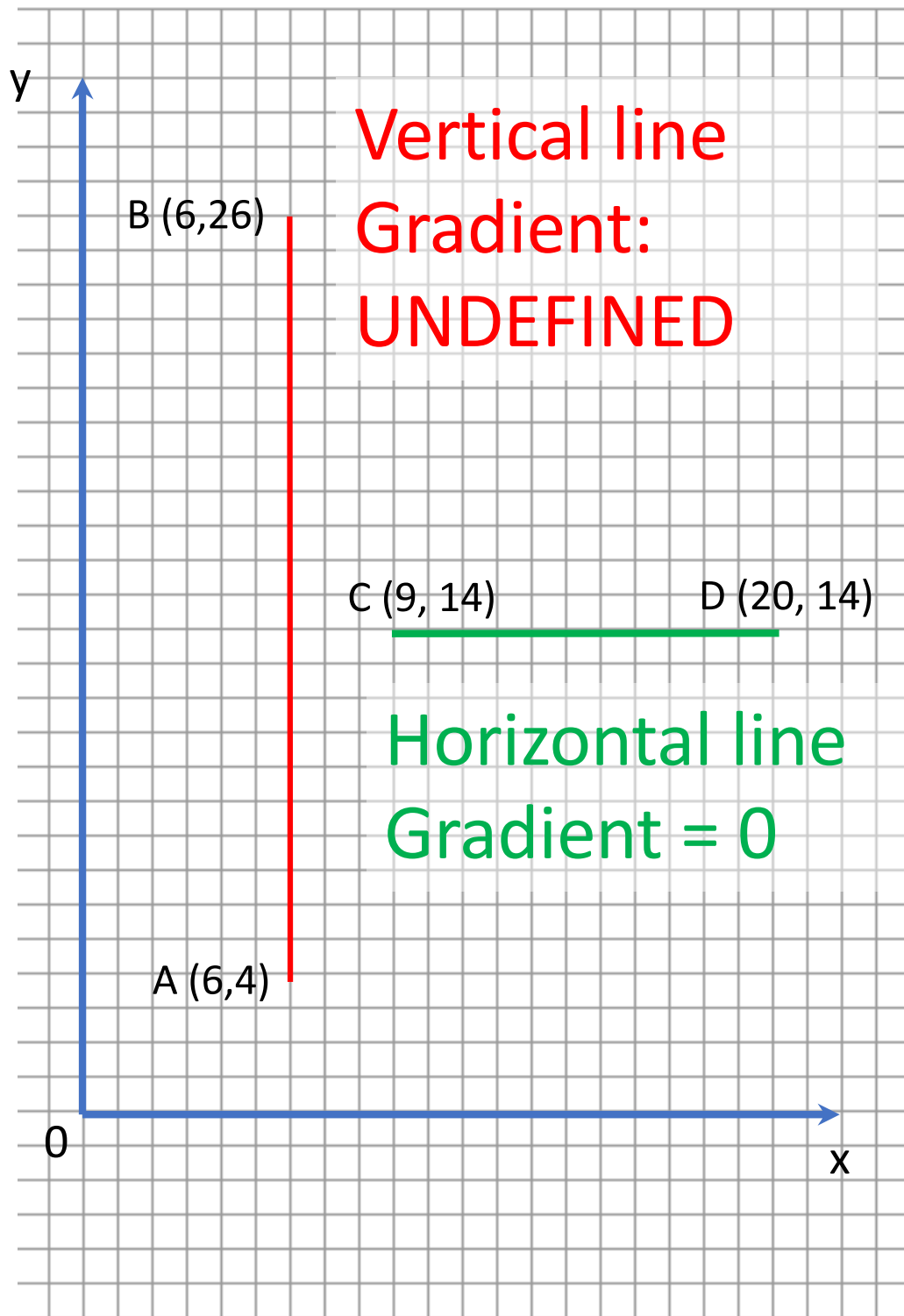
In general:

$$\text{Run} = x_2 - x_1$$

$$\text{Rise} = y_2 - y_1$$

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$(x_2 \neq x_1)$$



For VERTICAL line AB:

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{26 - 4}{6 - 6} = \frac{22}{0}$$

In this case, the run is 0, therefore, for a vertical line, there is no gradient.

For HORIZONTAL line CD:

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{14 - 14}{20 - 9} = \frac{0}{11}$$

In this case, the rise is 0, therefore, for a horizontal line, there is gradient = 0.

Examples

- Find the gradient of the interval joining the two points (7,4) and (10,28).

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{28 - 4}{10 - 7} = \frac{24}{3} = 8$$

- Find the gradient of the interval joining the two points (2,3) and (8,-7).

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-7 - 3}{8 - 2} = \frac{-10}{6} = -\frac{5}{3}$$